

Direct and indirect causes in the time series

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Introduction

The goal of this document is to describe the target function of the causal graph.

We treat the algorithm PCGCE in the paper *Extended summary graphs (ESG)* belongs to class of the exhaustive search algorithms. So to an optimal causal graph we make no difference between

- 1) exhaustive printing or combinatorial algorithms,
- 2) gradient-based algorithms including algorithms with relaxations,
- 3) zero-gradient algorithms.

The only thing we expect its result is a causal graph, optimal according to this criterion.

The model structure includes:

- 1) the time is continuous $\frac{dx}{dt} = f(x, t)$ (basic: discrete with $\Delta t = 1$),
- 2) the time series X, Y, Z has a spacial structure $\frac{dx}{dx} = g(x, s)$ (basic: scalar-valued),
- 3) there is no limitations of the time as in the ESG,
- 4) the model is linear in the latent space (basic: linear in the observable space),
- 5) the latent space represent a spectrum of the observable domain.

The graph structure:

- 1) the graph is a bipartite graph between the observable and the latent,
- 2) the direct graph is isomorphic to the reverse graph,
- 3) the causal graph is in the hypergraph that groups the variables or the time.

The time structure for the convolutional model $\int_{\tau=0}^T f(\tau)h(t-\tau)d\tau$

The causal graph $G = G(t)$ is a time series. We suppose either:

- 1) it is stationary for all t ,
- 2) it is stationary for $t \geq T$ for with the period T ,
- 3) it is non-stationary but with rejection to discover its structure.

1 Appendix. Natural Gradient Descent

The natural gradient descent is a method to optimize the model parameters θ by estimating the distribution

$$p(\theta) \propto \exp(-\mathcal{L}(\theta))$$

where $\mathcal{L}(\theta)$ is the loss function of the model. The natural gradient descent is defined as

$$\theta_{t+1} = \theta_t - \eta F^{-1}$$

where F is the Fisher information matrix of the model parameters θ .

Fisher Metric and Natural Gradient

When working with probabilistic models, we can use a metric that is more natural for probability distributions: the Kullback–Leibler (KL) divergence. Suppose we have a likelihood model

$$p(x \mid \theta),$$

where x denotes fixed data and θ is a parameter vector.

Let θ_t denote the current parameter values, and suppose we wish to find updated parameters θ_{t+1} . Under the natural gradient framework, this is done by solving the following optimization problem:

$$\theta_{t+1} = \arg \min_{\theta} [f(\theta_t) + \nabla f(\theta_t)^\top (\theta - \theta_t) + D_{\text{KL}}(p(x \mid \theta) \parallel p(x \mid \theta_t))].$$

In general, this minimization problem is intractable. However, the KL divergence can be approximated locally using a second-order Taylor expansion. This expansion yields the Fisher information matrix F , which captures the local curvature of the statistical manifold. In particular, around θ_t , we have

$$D_{\text{KL}}(p(x \mid \theta) \parallel p(x \mid \theta_t)) \approx \frac{1}{2}(\theta - \theta_t)^\top F(\theta - \theta_t),$$

where the Fisher information matrix is defined as

$$F = \mathbb{E}_{x \sim p(x \mid \theta)} \left[(\nabla_{\theta} \log p(x \mid \theta)) (\nabla_{\theta} \log p(x \mid \theta))^\top \right].$$

The Fisher information matrix therefore encodes the curvature of the likelihood-based loss function in parameter space. Using this approximation, the resulting natural gradient update rule becomes

$$\theta_{t+1} = \theta_t - \gamma F^{-1} \nabla f(\theta_t),$$

where $\gamma > 0$ is the learning rate.

Natural Gradient from an Information-Geometric Perspective

Consider a parametric family of probability distributions

$$\mathcal{M} = \{p(x \mid \theta) \mid \theta \in \Theta \subset \mathbb{R}^d\},$$

which forms a smooth statistical manifold $\mathcal{M}$. Each parameter vector θ corresponds to a point on this manifold, and the coordinate system θ is generally non-Euclidean.

The natural notion of distance between nearby probability distributions on $\mathcal{M}$ is given by the Kullback–Leibler divergence. For two infinitesimally close distributions $p(x \mid \theta)$ and $p(x \mid \theta + d\theta)$, the KL divergence admits the second-order expansion

$$D_{\text{KL}}(p(x \mid \theta + d\theta) \parallel p(x \mid \theta)) = \frac{1}{2} d\theta^\top G(\theta) d\theta + o(\|d\theta\|^2),$$

where

$$G(\theta) = \mathbb{E}_{x \sim p(x \mid \theta)} \left[(\nabla_\theta \log p(x \mid \theta)) (\nabla_\theta \log p(x \mid \theta))^\top \right]$$

is the **Fisher information metric tensor** on $\mathcal{M}$.

Let $f(\theta)$ be a smooth objective function defined on the manifold (e.g., the negative log-likelihood or empirical risk). Given the current point $\theta_t \in \mathcal{M}$, we seek a new point θ_{t+1} by minimizing a first-order approximation of f subject to a constraint on the geodesic distance induced by the Fisher metric. This leads to the optimization problem

$$\theta_{t+1} = \arg \min_{\theta} \{ f(\theta_t) + \nabla f(\theta_t)^\top (\theta - \theta_t) \mid D_{\text{KL}}(p(x \mid \theta) \parallel p(x \mid \theta_t)) \leq \epsilon \}.$$

Using the local quadratic approximation of the KL divergence, this constrained problem reduces to minimizing the linearized objective under a Riemannian norm constraint. The solution is given by a step in the direction of the **natural gradient**, defined as

$$\tilde{\nabla} f(\theta) = G(\theta)^{-1} \nabla f(\theta).$$

Accordingly, the natural gradient descent update takes the form

$$\theta_{t+1} = \theta_t - \gamma \tilde{\nabla} f(\theta_t) = \theta_t - \gamma G(\theta_t)^{-1} \nabla f(\theta_t),$$

where $\gamma > 0$ is the step size.

From an information-geometric standpoint, this update corresponds to moving along the direction of steepest descent with respect to the Fisher–Rao Riemannian metric, ensuring invariance under smooth reparameterizations of the statistical manifold.